



Technology-Driven Sports Digital Twin: Human-Machine Collaborative Olympic Strategic Decision System

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ABSTRACT

In the digital era, sports increasingly rely on intelligent technologies. This study proposes a strategic framework for the Olympic Games, combining statistical models, machine learning, and digital twin technology. Structured into data, model, and application layers, it fuses historical and real-time data for adaptive planning. Techniques like k-means clustering, time series forecasting, and Markov chains address cross-national differences, considering host nation and coach effects. A microservice-based digital twin improves prediction and system stability. Data augmentation and interpretable models enhance transparency. Results show host nations gain 9.52% more medals, while middle-income countries benefit from targeted training and efficient strategies. The framework forecasts the 2028 medal distribution.

SYSTEM CONSTRUCTION

The research divides into two phases:

- 1) Foundational prediction and strategy exploration
- 2) Multi-model integration and digital twin platform

The following sections outline the methodologies and system architecture across each phase, focusing on the data and model layers, with selective attention to the application layer.

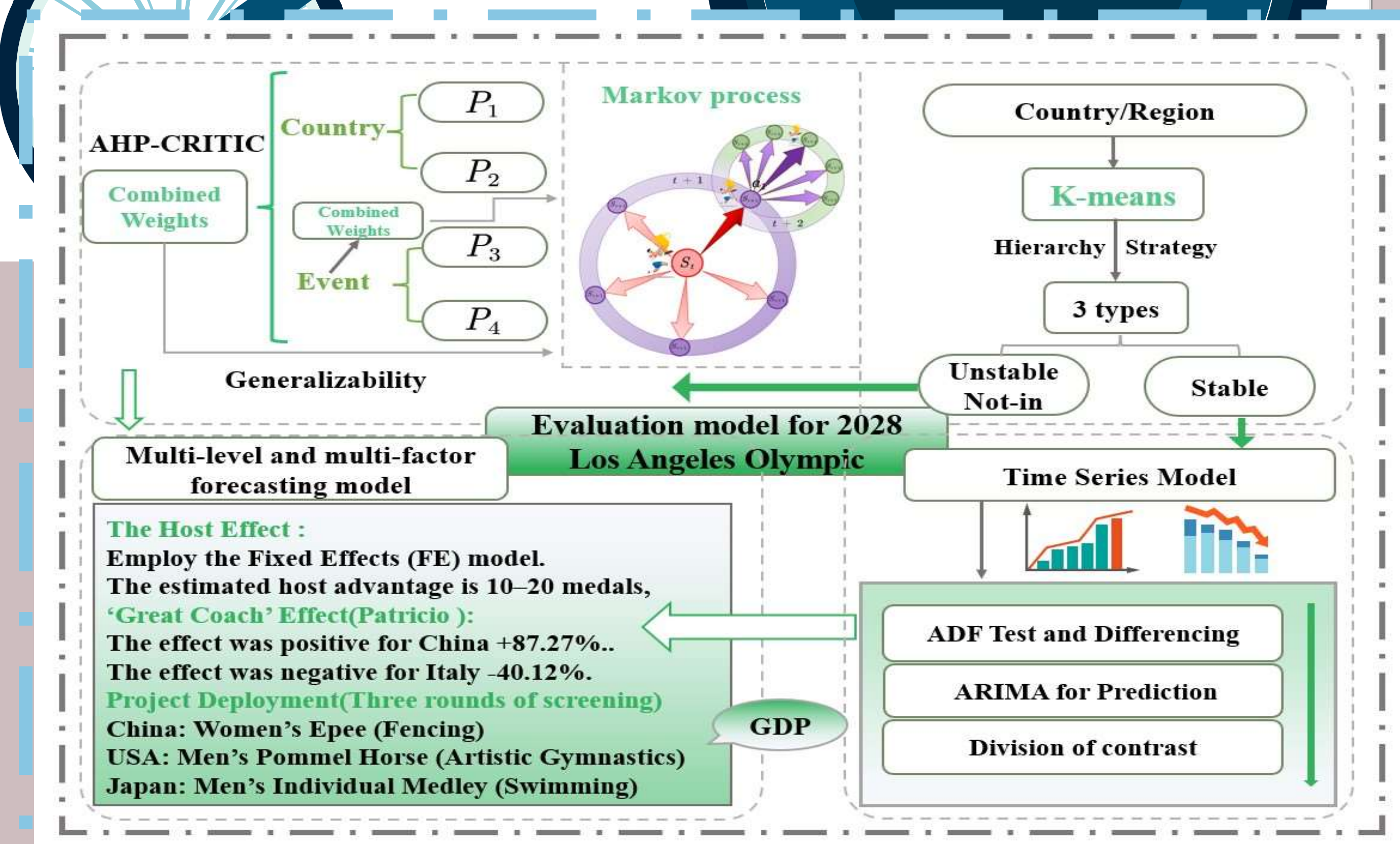


Figure 1. The structure of phase 1.

We propose a hierarchical prediction framework, using standardized Olympic data (athletes, hosts, medals, events) with consistent national codes for fair comparison.

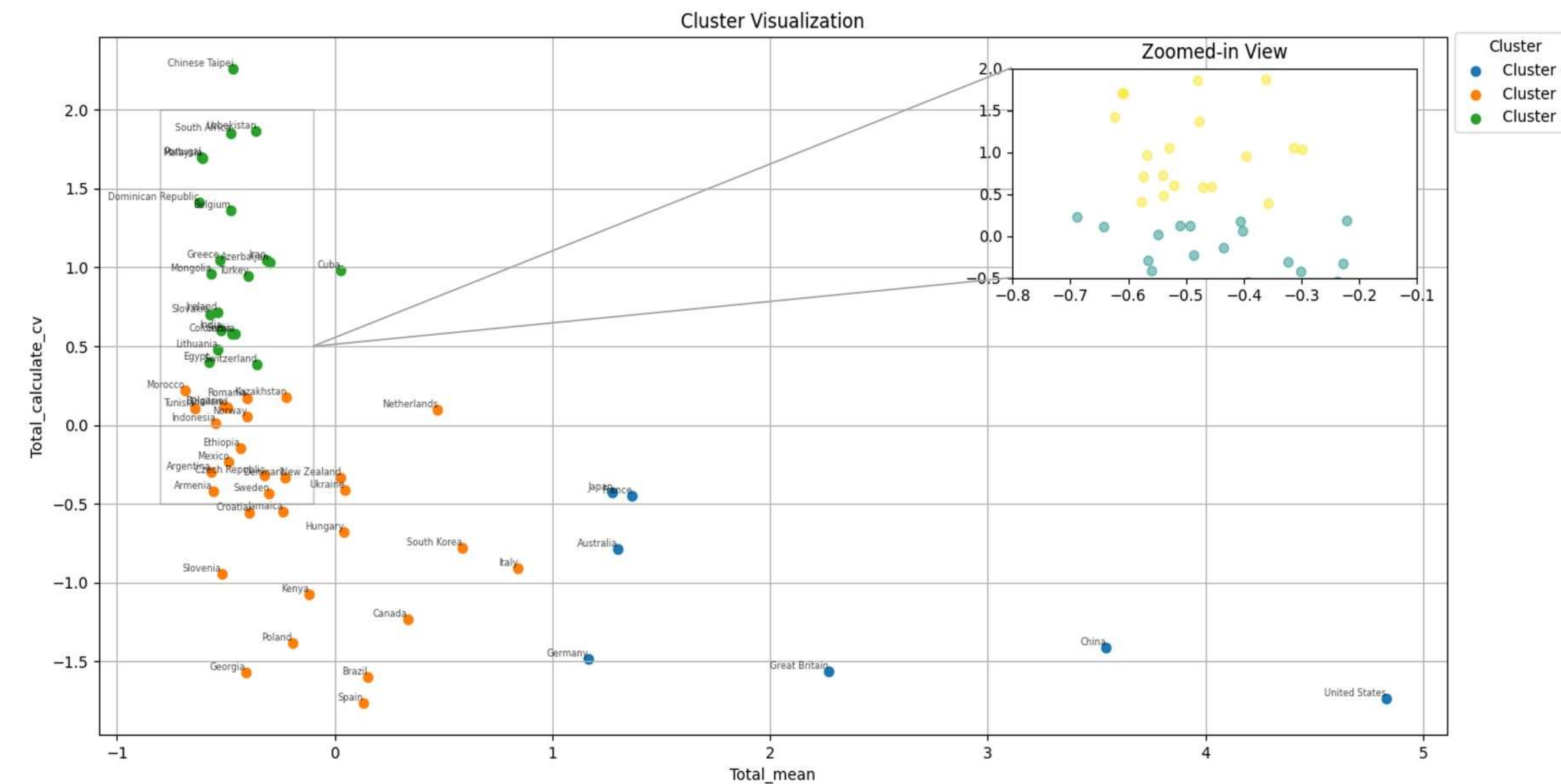


Figure 2. K-means Clustering Visualization.

Phase 2 integrates real-time IoT and social data, combining models in a microservice digital twin. Predictions are fused with optimized weights, using stacking and boosting to enhance accuracy.

- ✖ The data layer aggregates historical and real-time data via ETL pipelines and APIs.
- ✖ The model layer hosts scalable microservices.
- ✖ The visualization layer offers dashboards for scenario analysis and model outputs.

We also consider key influencing factors:

Host nations usually win more medals due to better infrastructure and investment, with hosting boosting counts by 9.52% (estimated by Fixed Effects and LSDV models).

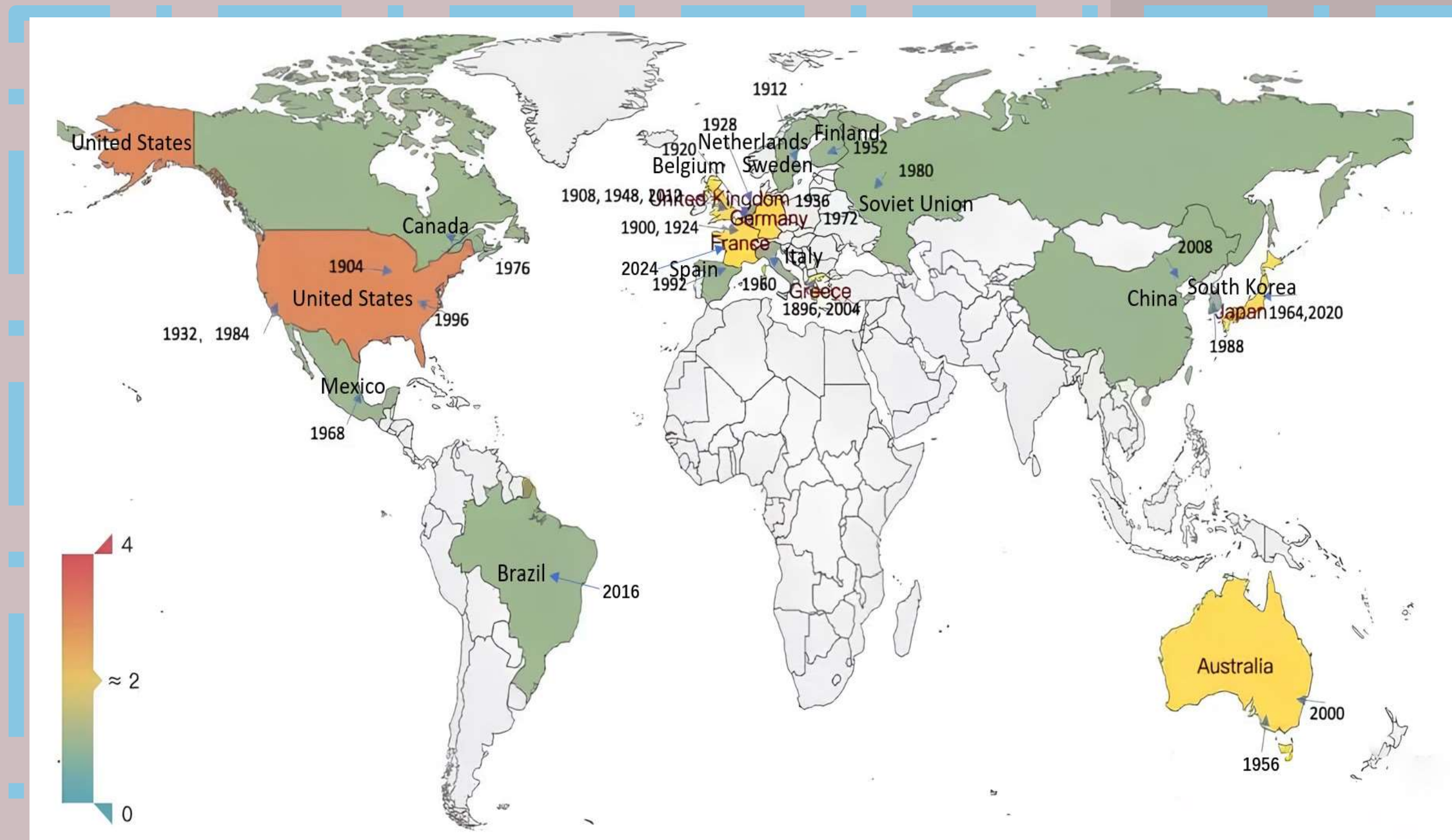


Figure 3. Distribution of previous Olympic host countries.

Elite coaches moving between countries also impact results, as seen when Italy's coach Patricio joined China in 2020. These effects guide strategy recommendations.

ETHICS & FAIRNESS GUARANTEE

We enhance fairness by using SMOTE to balance small-country data and focusing on event-level results for limited cases.

Bias penalties and fairness metrics (Equal Opportunity, Equalized Odds) ensure equal treatment.

LIME and SHAP improve transparency by explaining predictions and feature impacts. Together, these steps make our system fair, clear, and globally inclusive.

MEDAL PREDICTION AND FAIRNESS

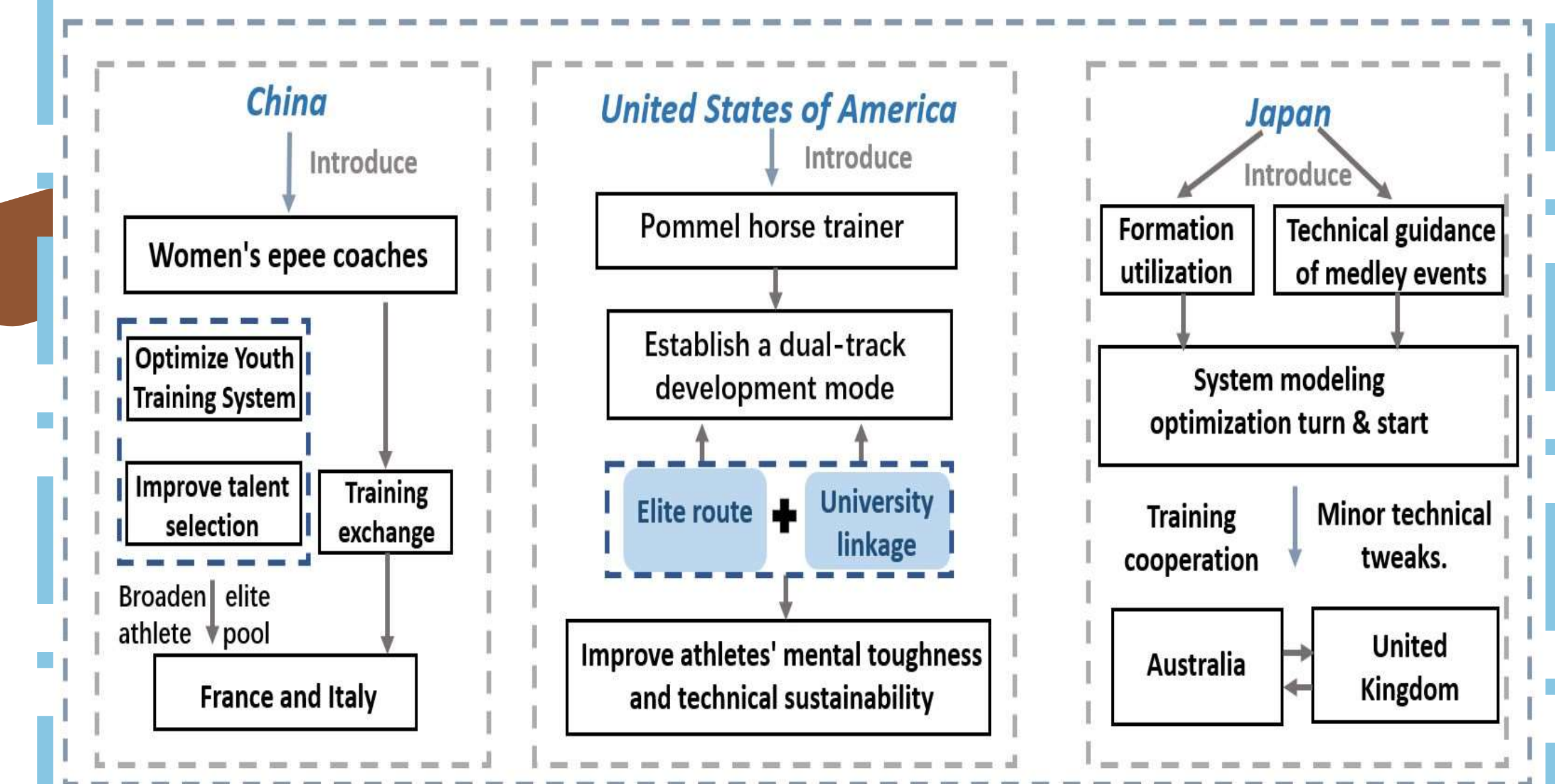


Figure 4. Strategic deployment proposals for 3 pilot cities.

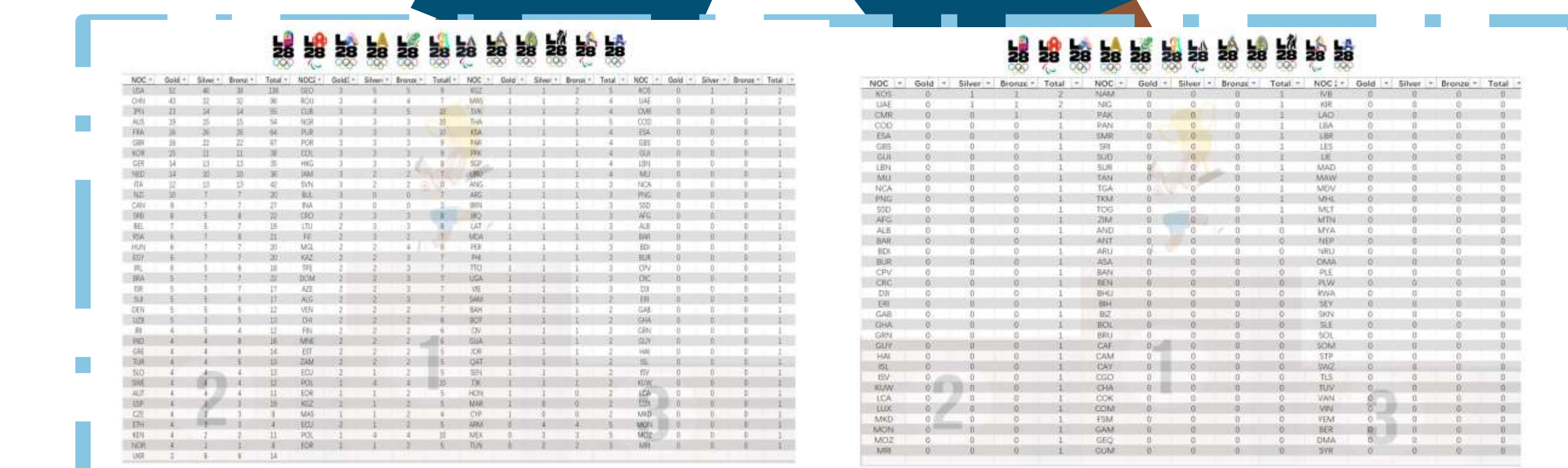


Figure 5. Medal List

Medals rise with GDP and population but level off. Big nations like China and the USA benefit; India lags without strong infrastructure. Hosts win more due to resources, and investment keeps their lead. Poorer countries target few sports; richer ones compete broadly.

CONCLUSION & FUTURE WORK

This study creates an Olympic decision system using machine learning and digital twins for accurate medal predictions. It ensures fairness with SMOTE, LIME, and SHAP.

Future Work: Adding real-time data and adaptive models for accuracy and fairness.